ADMINISTRATIVE NOTICE

NOT MEASUREMENT SENSITIVE

JOINT-GRA-003
NOTICE 1
1 August 2022

US AIR FORCE MODULAR OPEN SYSTEMS APPROACH (MOSA) PORTFOLIO – ALL SOURCE POSITION AND NAVIGATION (ASPN), BIG IRON, COMMON OPEN ARCHTIECTURE RADAR PROGRAMS (COARPs), FUTURE AIRBORNE CAPABILITY ENVIRONMENT™ OR FACE™, OPEN MISSION SYSTEMS (OMS), RED HAWK, SOFTWARE COMMUNICATION ARCHITECTURE (SCA), SENSOR OPEN SYSTEMS ARCHITECTURE OR SOSA™, SPACE SYSTEMS COMMAND (SSC) COMMON PAYLOAD INTERFACE STANDARD (CoPaIS) UNIVERSAL ARMAMENT INTERFACE (UAI), UNIVERSAL COMMAND AND CONTROL INTERFACE (UCI), WEAPON OPEN SYSTEMS ARCHITECTURE (WOSA),

MODULAR OPEN SYSTEMS APPROACH (MOSA) STANDARDS ADOPTION

The U.S. Air Force worked for over a decade to develop non-government standards, and create government standards when non-government standards are not available to support MOSA strategies. As the Air Force seeks to mature Government Reference Architectures (GRAs) in support of an overall MOSA strategy, the standards the Service uses must mature along with the reference architectures. The following (government and non-government) MOSA enabling standards demonstrate applications for Weapon Systems being developed by Air Force program Offices: All Source Position and Navigation (ASPN), Big Iron, Common Open Architecture Radar Programs (COARPs), Future Airborne Capabilities Environment or FACE, Sensor Open Systems Architecture or SOSA, Open Mission Systems (OMS), RedHawk, Software Communication Architecture (SCA), Universal Command and Control Interface (UCI), Universal Armament Interface (UAI), Weapon Open Systems Architecture (WOSA).

STANDARD SPECIFICATION

ALL SOURCE POSITION AND NAVIGATION (ASPN)

ASPN developed a standardized sensor integration system used by multiple Services incorporating non-traditional sensor sources to augment Position Navigation and Timing (PNT) in Global Positioning System denied environments. An Interface Control Document (ICD) Protocol Specification provides access to navigation information no matter the source, sensors and software producing the information. The Air Force is feeding ASPN ICD information to programs working on PNT Reference Architectures.

The ASPN ICD will be available on the Defines Intelligence Information Enterprise (DI2E) at https://bitbucket.di2e.net/projects/ASPN/repos/aspn-icd/browse. However, DI2E is scheduled to sunset in 2022. After the sunset of DI2E the information will transfer to https://aspn.us.

BIG IRON

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:30Z
Check the source to verify that this is the current version before use.

Big Iron is an Air Force sponsored multi-phase effort to design and implement a non-proprietary, modular and open architecture that incorporates software driven solutions for next generation electronic warfare (EW) systems. Currently the Department of the Air Force (DAF) is the only Service using the Big Iron standard. The Big Iron standard is being designed to leverage Open Mission Systems, and other EW schema elements to enable rapid integration and flexibility to support platform interfaces, data sharing, and portability across a breadth of adopting programs.

The Big Iron standard is located on the Defense Intelligence Information System https://di2e.net.

STANDARD SPECIFICATION

COMMON OPEN ARCHTIECTURE RADAR PROGRAMS (COARPs) – AIR FORCE LIFECYCLE MANAGEMENT CENTER SENSORS DIRECTORATE

COARPs is a Government led standard driven by a community of members with the goal of diving radar systems into separately procurable subsystems. Currently the DAF is the only Service using Big Iron. The COARPs standard seeks to enable independent computing hardware refresh without major software rewrite, and looks to insert third party mode technology into radar processing subsystems. The COARPs standard is evolving into a multi-intelligence architecture to achieve the following objectives:

- Sensor-specific managers to control individual sensors
- Multi-intelligence coordinator supporting data fusion and cross cueing
- Modality specific front-end interfaces to provide flexibility and performance
- Application software to leverage sensor functions

The COARPS standard is located on the Air Force Virtual Distribution Library https://www.vdl.afrl.af.mil.

STANDARD SPECIFICATION

FUTURE AIRBORNE CAPABILITY ENVIRONMENT™ – THE OPEN GROUP FACE CONSORTIUM

The Open Group FACE Consortium is a Government and industry partnership, which also includes academia participation, to define an open avionics environment for all military airborne platform types. The Army, Air Force, and Navy Participate on the FACE Consortium. The FACE Consortium works to develop and consolidate the open standards, best practices, guidance documents, and business models to achieve the following objectives:

- Standardized approaches for using open standards for software development and integration
- Lower implementation costs of FACE systems
- Standards that support a robust architecture and enable quality software development
- The use of standard interfaces that will lead to the reuse of capabilities
- Portability of applications across multiple FACE systems and vendors
- Procurement of FACE conformant products
- More capabilities reaching the warfighter faster
- Innovation and competition within Department of Defense (DoD) acquisition
- Robust, interoperable, portable, and secure computing platforms

More information regarding the FACE standard can be found via https://www.opengroup.org/face.

STANDARD SPECIFICATION

OPEN MISSION SYSTEMS (OMS)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:30Z
Check the source to verify that this is the current version before use.

The goal of the OMS is to develop industry and DoD consensus for a non-proprietary mission system standard that enables affordable technical refresh and insertion, simplified mission systems integration, service reuseand interoperability, and competition across the life cycle. Industry and the Government worked on solutions cooperatively to achieve these goals for several years with extensive testing and demonstrations. The OMS Standard is being used by the DAF and Navy.

The OMS standard can be found via https://www.vdl.afrl.af.mil/programs/oam/oms.php.

STANDARD SPECIFICATION

RED HAWK

RedHawk is a software-defined radio (SDR) standard designed to support the development, deployment, and management of real-time software radio applications. To support the design and development of software applications, RedHawk provides tools that allow development and testing of software modules called "Components" and composition of Components into "Waveform Applications" that can be seamlessly deployed on a single computer or multiple network-enabled computers.

The RedHawk integrated development environment (IDE) provides tools to support development of RedHawk software. The development and deployment of RedHawk Applications are aided by graphical editors and drag-and-drop Waveform construction. The IDE allows users to interact with and control multiple running RedHawk instances and applications.
The RedHawk standard can be found via https://RedHawksdr.org.

STANDARD SPECIFICATION

SOFTWARE COMMUNICATION ARCHITECTURE (SCA)

The Software Communications Architecture (SCA) is published by the Joint Tactical Networking Center (JTNC). This architecture was developed to assist in the development of Software Defined Radio (SDR) communication systems, capturing the benefits of recent technology advances which are expected to greatly enhance interoperability of communication systems and reduce development and deployment costs.
The SCA standard is available via https://www.jtnc.mil/Resources-Catalog/Category/16990/sca/.

STANDARD SPECIFICATION

SENSOR OPEN SYSTEMS ARCHITECTURE™

The Open Group SOSA Consortium enables Government and industry to collaboratively develop open standards and best practices to enable, enhance, and accelerate the deployment of affordable, capable, and interoperable sensorsystems. The Army, Air Force, and Navy Participate on the SOSA Consortium. The SOSA Consortium is creating open system reference architectures applicable to military and commercial sensor systems and a business model that balances stakeholder interests. The architectures employ modular design and use widely supported, consensus-based, nonproprietary standards for key interfaces that are expected to:

- Reduce development cycle time and cost
- Reduce systems integration cost and risk
- Increase commonality and reuse
- Reduce sustainment and modernization cost
- Support capability evolution and mitigate obsolescence
- Enable technology transition

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:30Z
Check the source to verify that this is the current version before use.

- Facilitate interoperability
- Isolate the effects of change

More information regarding the SOSA aligned/upcoming standard can be found via https://www.opengroup.org/sosa.

STANDARD SPECIFICATION

SPACE SYSTEMS COMMAND (SSC) COMMON PAYLOAD INTERFACE STANDARD (CoPaIS)

The Common Payload Interface Standard (CoPaIS) standardizes the space vehicle bus-to-payload interface. It is being developed incrementally starting with the command and data handling (C&DH) portion of the standard (SMC-SIS-001) that was released in July 2020 as the first United States Space Force interface standard. As part of the SSC Enterprise Standards Library, it is recommended for use by all SSC programs. While SMC-SIS-001 support both ground-based and on-orbit bus-to-payload integration, the other portions (e.g., mechanical, thermal, etc.) are specifically for on-orbit bus-to-payload integration. The CoPaIS payload can be small to large mission payloads on the nadir side of the bus or a modular propulsion subsystem on the zenith side of the bus. The non-C&DH portions of the standard will be matured in in an IEEE-ISTO Space Systems Command MOSA Interface Standards Alliance subcommittee.

SMC-SIS-001 can be provided upon request from the SSC/BZEE organization. Information and requests for participation in the SSC MOSA Interface Standards Alliance are available at http://InteroperabilityAlliance.space.

STANDARD SPECIFICATION

UNIVERSAL ARMAMENT INTERFACE (UAI)

UAI is a DoD initiative to develop standardized functional interface in aircraft, weapon and mission planning to support integration of future air-to-surface weapons independent of aircraft Operational Flight Program cycles. Currently the DAF is the only Service using UAI. UAI applies to air-to-ground weapons and platforms that are MIL-STD 1760 compliant and is comprised of three parts:

- Platform to Store interface
- Mission Planning Interface
- Launch Acceptability Region standard

For Access to UAI Materials please contact the program office at AFLCMC.WWX.UAIProgramOffice@us.af.mil.

STANDARD SPECIFICATION

UNIVERSAL COMMAND AND CONTROL INTERFACE (UCI)

Universal C2 Interface (UCI) is a messaging standard supporting machine-to-machine communication for mission-level command and control (C2). UCI has some similarities to the Unmanned Systems Control Segment (UCS) standard, however more information on UCS is available in Administrative Notice JOINT-UXS-001. UCI defines a set of messages intended to standardize interfaces between software services for C2, and these messages have been defined and validated as fit for purpose across a variety of mission behaviors and heterogeneous mission system implementations. The UCI Standard is being used by the Department of the Air Force and Navy, but the Air Force manages the Collaborative Working Group of Industry members who manage and update the UCI standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:30Z
Check the source to verify that this is the current version before use.

The UCI standard can be found via https://www.vdl.afrl.af.mil/programs/oam/.

STANDARD SPECIFICATION

WEAPON OPEN SYSTEMS ARCHITECTURE (WOSA)

Weapon Open System Architecture (WOSA) is a standard for munitions, fostering modular design of weapon systems, developed, and maintained by the Munitions Open Architecture Test and Evaluation Laboratory (MOATEL) team. Currently the DAF and Navy are using WOSA. The WOSA standard is implemented in the earliest phases of acquisition, from acquisition strategy and maintained through a munition's entire lifecycle. MOATEL performs specification updates, contract language and guidance, requirements development and analysis, documentation validation, tool development, system and component evaluation, and support to program offices to implement WOSA on a program. WOSA is currently a separate effort from UAI, but the two are looking to combine into a single standard in late 2022.

The WOSA standard can be found on the All Partners Access Network: https://community.apan.org/wg/wosa/.

CONCLUDING MATERIAL

Custodian:

Air Force – 01

Navy - SH

Preparing Activity:

Air Force – 01

Review activities:

Air Force – 10, 19

Navy – AS, OS

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:30Z
Check the source to verify that this is the current version before use.